

Learned Model Probability Cohort Analysis

Statbirt | Historical final decisions through 2026-05-22 | Prepared 2026-05-25

Headline: The search found several learned-probability 75-80% cohorts with at least 100 decisions and hit rates above 90%. The strongest verified slice was 94 hits in 101 decisions, a 93.1% hit rate.

Source And Baseline

The analysis joined learned model probabilities from **model_predictions.csv** to candidate-side final results in **statbirt_candidates.csv** on date, player_id, and game_pk. Only final decisions with a resolved hit/no-hit result were counted.

Population	Hits / Decisions	Hit Rate
All final candidate decisions	32,678 / 57,750	56.6%
All learned 75-80% probability picks	329 / 432	76.2%

Strict 75-80% Cohorts Above 90%

Cohort	Hits / Decisions	Hit Rate
hitter_ba_25_ab <= .300 + pitcher_hpi_200 >= 1.00 + pitcher_hpi_season >= 1.10	94 / 101	93.1%
hitter_bb_rate_season >= .080 + hitter_last_5_games_ba <= .400 + pitcher_lr_opp_ba_50 >= .260	103 / 113	91.2%
hitter_ba_25_ab <= .350 + hitter_bb_rate_season >= .080 + pitcher_lr_opp_ba_50 >= .260	91 / 100	91.0%
hitter_ba_25_ab <= .300 + pitcher_hits_last_18_ip >= 18 + pitcher_lr_opp_ba_200 >= .260	96 / 106	90.6%
hitter_ba_2500_ab >= .280 + hitter_bb_rate_500_pa >= .060 + pitcher_hpi_season >= 1.10	101 / 112	90.2%

Original 51/54 Signal

The original high-signal slice also verified: learned probability of 75-80%, congregation member, and exactly 1-2 stop valves produced 51 hits in 54 resolved decisions, a 94.4% hit rate. It did not broaden cleanly by itself.

Expansion	Hits / Decisions	Hit Rate
Original: congregation + exactly 1-2 stop valves	51 / 54	94.4%
Add 0-stop players	56 / 60	93.3%
Add 3-stop players	73 / 82	89.0%
All players, exactly 1-2 stop valves	85 / 104	81.7%

Expansion	Hits / Decisions	Hit Rate
Congregation, 72-80% probability, exactly 1-2 stop valves	142 / 168	84.5%

Interpretation

The strongest broadened signal appears to be: **good learned probability + recently under-hot hitter + vulnerable starter.**

The best cohort is especially interesting because **hitter_ba_25_ab <= .300** sounds counterintuitive at first. Paired with weak starter HPI, it may be catching good hitters who are not overheated but are in favorable contact matchups.

Caveat

Use carefully: These cohorts were mined from historical data, so they should be treated as high-priority hypotheses rather than guaranteed edges. The 94/101 cohort has a Wilson 95% interval of roughly 86.4% to 96.6%. The next useful step would be walk-forward testing.